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# An Analysis of Data Mining Techniques in Software Engineering Database Design

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## Abstract

With the advent of Internet information technology, data mining technology has been applied in various fields of China's social and industrial development, and has promoted the quality development of the industry [1]. Nowadays, people are widely influenced by Internet computer technology, and the application of computer technology has been indispensable in life, work and study. At the same time, data mining technology arises from Internet communication technology and is used as an important technical means of operation and development by various industries, especially the application of data mining technology in university software engineering teaching is becoming more and more widespread, but there are still many unavoidable problems that require scholars to pay more attention. The article analyzes and discusses the significance of applying data mining technology in software engineering [2], and on this basis, the application path of data mining technology in software engineering database design is proposed, including: mining information, mining loopholes, mining execution records and open source. It is hoped that relevant technology users can efficiently apply data mining technology to make software engineering. It is more reasonable and efficient, so that software engineering can be further developed. This paper presents the implementation and performance analysis of the teacher management decision-making system based on data mining, and introduces the implementation process of the system in detail.

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**Keywords:** Analysis; Data Mining Technology; Database; Design; Software Engineering

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## 1. Introduction

With the rapid development of network technology, the demand for software is also gradually increasing, but due to the increasing scale of software systems, as well as more complex, prompting software development is more

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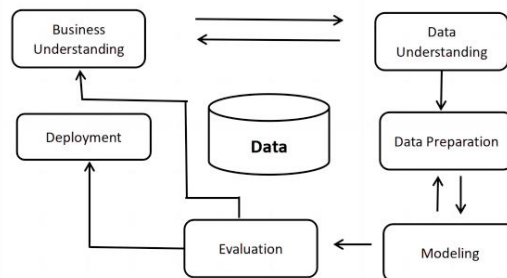
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difficult to control. Data mining techniques are gradually being applied in software engineering to solve the difficult data problems that exist in traditional software development methods, as well as the problems caused by the explosive growth in the amount of information. Its main purpose is to extract data with special value information from a large amount of data through the application of technical means. The application in software engineering is to find out the data information with value in the software engineering database, which can effectively locate system faults, predict the loopholes in the system, conduct cost assessment, etc. The role of data mining technology is to extract and analyze a large amount of information data, highly targeted, to carry out software engineering development, cannot be separated from the application of data mining technology, therefore, the study of how to apply data mining technology in software engineering on data can be designed is an important path to achieve this paper to improve the effectiveness of software development.

## 2. Introduction to Data Mining Techniques

From the point of view of data mining technology itself, it mainly refers to a modern information processing technology, and compared with traditional information processing technology, data mining technology is widely used in various fields with its own powerful functions. At present, data mining technology has not been fully popularized and promoted, and most IT industries lack in-depth understanding of the value of data mining technology applications and still use traditional information processing technology, which makes information processing efficiency relatively low. Usually, data mining technology contains data processing, data transformation, data analysis and other functions, and there is a certain correlation between each function to achieve a comprehensive assessment. For some software engineering enterprises, by applying data mining technology to software engineering, it can effectively improve the efficiency of information processing and prevent unnecessary errors from occurring, which plays an important role in the stable development of enterprises. Data mining process is shown in Figure 1.



**Fig. 1.** Data mining flow chart

Data mining techniques were first referred to as knowledge discovery in databases and were developed on the basis of databases. With continuous development, it gradually transforms from theoretical research to practical application and interoperates with other fields of science. The concept of software engineering first emerged in 1967. Software engineering focuses on maintaining and building high-quality, practical and effective software through an engineering approach, with the goal of developing software products for customers that meet requirements under established schedule and cost conditions. The application of data mining in engineering software began at the end of the twentieth century, proposed by Allen and other related scholars, the application of data mining to find out the reuse relationship in the code, since then, data mining technology in software engineering, to continue to apply and develop [3].

### 3. Common Approaches to Data Mining Techniques

#### 3.1 Neural Network Approach

This approach mainly simulates the neurons of the human brain and uses Hebb's learning rules and the MP model as an important basis, while representing the knowledge in terms of the connection weights of the neural network, where the learning is mainly reflected in the step-by-step calculation of the corresponding weights. In this approach, the most commonly used models include self-organizing networks, feedforward networks, and feedback networks, which are used for clustering, prediction, and associative memory, respectively.

#### 3.2 Genetic Algorithm Applications

This approach is based on natural biological screening and genetically motivated random search algorithms, and is one of the most common modes of operation in data mining techniques. In the application of genetic algorithms, the greatest advantage lies in the ability to make connections between models of many apparently unrelated things and to ensure that they develop in a synergistic and comprehensive way. At the same time, the increasing emphasis on genetics in biology has made it possible to use this method more effectively.

#### 3.3 Visualization Technology Application

This technique is mainly used to present the data with the help of certain graphs, for example simple graphical presentation of the data in the database, which can be presented visually to the researcher visualize monotonous data and make it easier for researchers to The information can be comprehensively grasped.

#### 3.4 Statistical Analysis Method Application

In the application of this method, technicians usually use principal component analysis and regression analysis to comprehensively analyze the relationships between fields of large databases, of which the relationships mainly cover multiple relationships, but the functional relationship is one of the most important relationships. The various functional relationships can often be expressed in the form of formulas, but the related relationships are difficult to be expressed by some specific formulas, and the application of statistical analysis can make reasonable use of such relationships. The application of statistical analysis can make reasonable use of such relations.

#### 3.5 Decision Tree Technology Applications

This technology is often applied to classify and integrate various types of information data, especially when handling large amount of data and information, it not only helps researchers to find the key information in a short period of time, but also to simplify and speed up the classification of information, while ensuring It can also simplify and speed up the classification of information and ensure more accurate classification. However, there are some drawbacks of this technology, mainly in its noise immunity performance. The main drawbacks of this technique are its relatively poor noise immunity and its inability to represent more complex ideas in a comprehensive manner.

### 4. Application of Data Mining Technology in Teaching Management Database

#### 4.1 Achievement Classification Model Building

The grade classification model can be obtained by the following steps: On the basis of the sample data records of the information entered in the student grade database of the teaching system, a decision tree algorithm is selected to build a decision tree that can describe the characteristics of student grades, and a student grade classification model is constructed as shown in Figure 2.

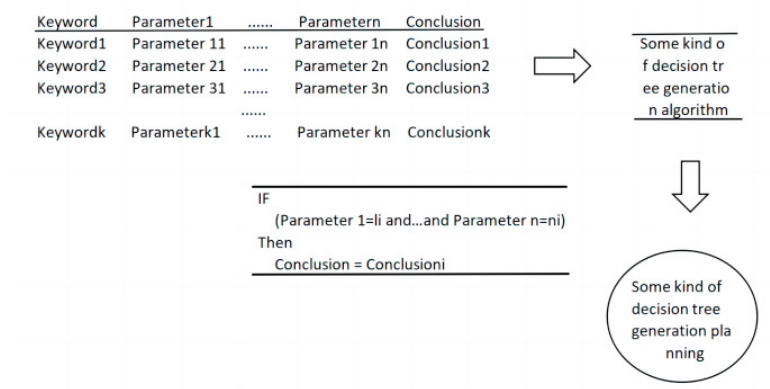


Fig. 2. Building a tree

#### 4.2 Student Performance Assessment

Using the model constructed in the previous step, a conditional rule is generated and used to make a prediction about the performance of the students taking the test as shown in Figure 3.

Specifically, the workflow for assessing student achievement is as follows: (1) summarize the characteristic attributes of student achievement; (2) analyze and process the characteristic attributes of student achievement; (3) design a classification model for assessing student achievement based on the above achievement summarization and analysis; (4) input student achievement and make classification prediction of achievement.

In the student performance database of teaching management, student performance has many performance attributes, such as class, student number, major, grade, etc. First, analyze and summarize these attributes. The purpose of this action is to eliminate irrelevant content or weakly relevant content content, and select the content that is positively related to the target as the node of the decision tree from a large number of attributes [4]. The basic method of attribute aggregation is to use the query function in the database to collect the data and information of the target content, and present the results to the user in the form of Figure 3 and rules below.

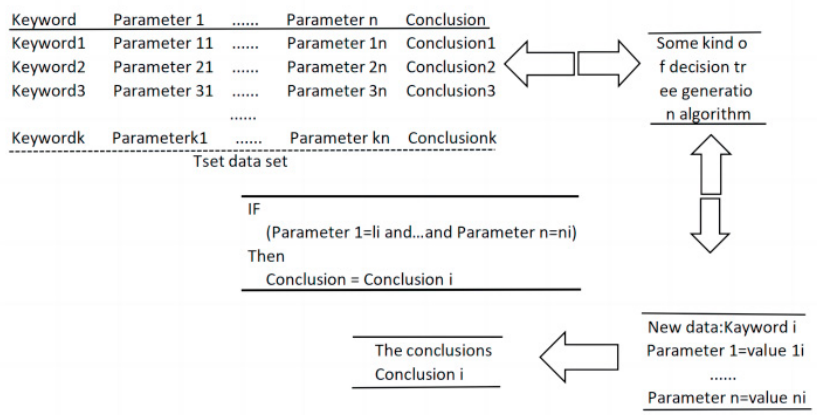


Fig. 3. Prediction

Decision trees, also known as classification trees, are one of the classification methods in data mining that can perform category or continuous type variables for classification prediction. The concept of decision tree is to divide the data into different categories by the known characteristics of the data or classification items, and each category then has a different decision pattern. In the process of decision tree generation, according to the information gain, the decision tree looks for the variable with the maximum information in the data set to establish data nodes, and

then establishes branches according to the characteristics of the variables, and each branch subset repeats the process of establishing branches and nodes until the completion of the whole decision tree, and each path of the decision tree represents a classification rule, and the new data is predicted by this classification rule [5].

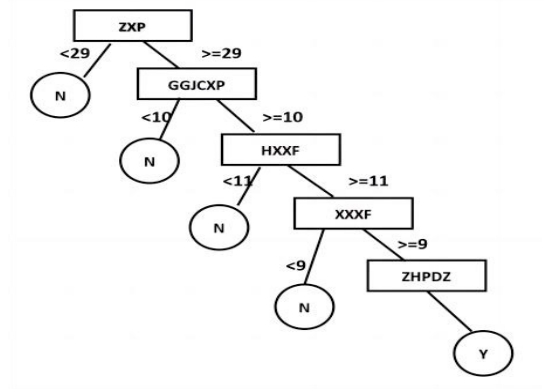


Fig. 4. Decision tree of whether to attend the graduation defense on schedule

A decision tree was built using the achievement pool of 200 students enrolled in a grade as the training pool and 6 attributes such as Zxf, Ggjcif, Hxxf, Xxxf, Zhpdz, Khjg as the examination index [6]. Among the existing decision tree classification methods, ID3, C4.5, classification and regression trees are the most widely used, and these three methods are suitable for textual symbols or discrete state data type, Quinlan improved the traditional C4.5 method in 1996. It makes the processing of continuous variables more efficient. Decision tree is a supervised machine learning method, and this process resembles a tree structure, and the decision tree generated using ID3 is shown in Figure 4 and Figure 5.

The specific code is shown below:

```

if (optarg && *s) { /* oplarg=s; optarg=NULL: break; */
    }
    /* get option argument */
    Else {
        /*--if argument is no option */
        switch(k++){
            /*evaluate non-options */
            case 0: fn_in =s; break;
            case 1: fn_out =s; break;
            case 2: fn_app =s; break;
            default: error(E_ARGCNT); break;
        }
        /* note filenames */
    }
}

If(optarg) error(E_OPTARG); /*check option argument */
if ((k<2) || (k>3)) /* and the number of arguments */
    error(E_ARGCNT); /* (either in/out or in/out/app) */
if ((!fn_in || !*fn_in) && (fn_app && !*fn_app))
    error(E_STDIN); /* stdin must not be used twice */
Switch(target) {
    /* check and translate target type */
    case 's': target = TT_SET; break;
    case 'm': target = TT_MFSET; break;
    case 'c': target = TT_CLSET; break;
    case 'r': target = TT_RULE; break;
    case 'h': target = TT_HEDGE; break;
    default: error(E_TARGET, (char)target); break;
}

if (supp>1)
    error(E_SUPP, supp); /*(< 0: absolute number) */
if ((conf < 0) || (conf > 1))
    error(E_CONF, conf); /*check the minimal confidence */
if (minlen <= 0) error(E_RULELEN, minlen); /* check the limits */
if (maxlen <= 0) error(E_RULELEN, maxlen); /* for the rule length */

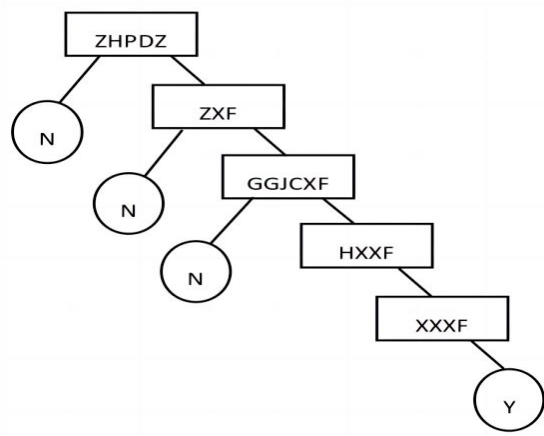
```

```

switch (arem){
    /* check and translbte measure */
    case 0: case '0': arem = EM_NONE; break;
    case 'd': case '1': arem = EM_DIFF; break;
    case 'q': case '2': arem = EM_QUOT; break;
    case 'a': case '3': arem = EM_AIMP; break;
    case 'i': case '4': arem = EM_INFO; break;
    case 'c': case '5': arem = EM_CHI2; break;
    Default : error(E_MEASURE,(char)arem); break;
}

if ((minval<0)
    || ((target== TT_RULE)&&(arem!= EM_AIMP)&&(minval>1)))
    error(E_MVAL,minval); /*(must usually be between 0 and 1) */
if (target==TT_HEDGE){ /* in hyperedge mode */
    minval = conf; conf = 1;} /* ada pt the parameters */
else if (target <= TT_CLSET){ /* in item set mode neutralize*/
    rsdef=IST_BOTH; conf = 1;} /* rule specific settings */
if (arem ==EM_NONE) /* if no add. rule eval. Measure,*/
    aval = 0; /* c lear the corresp.output flag */
if ((filter<=-1) || (filter>= 1))filter=0;

```



**Fig. 5.** Illustration of the decision tree for the possibility of early degree study

Teacher and student related data analysis module implementation. The algorithm selection for the teacher-student related assessment module uses an improved decision tree ID3 algorithm, which builds a decision tree based on the information in the student performance database and then analyzes the students for assessment [7]. The following is a description of the initialization and variables of the main variables of the ID3 and C4.5 decision tree programs, and the key code to implement them. When the user clicks on the "Decision Tree ID3 Data Mining" menu item in the main interface menu, student assessment is performed. The algorithm chosen for the student assessment module is the modified Decision Tree ID3 algorithm, which builds a decision tree based on the information in the student achievement database [8]. Students are then analyzed and evaluated. The algorithm chosen for the teacher-student related data analysis module is the improved Apriori algorithm, which takes the student performance database, the basic personal information of teachers and students, and the information of students' evaluation of teachers as the data mining objects, explores the relationship between each influencing factor, discovers the laws, and provides information support to teaching management decisions [9]. The ID3 decision tree method can also be extended to the fuzzy ID3 based on fuzzy set theory decision tree. The flow chart of ID3 algorithm is shown in Figure 6.

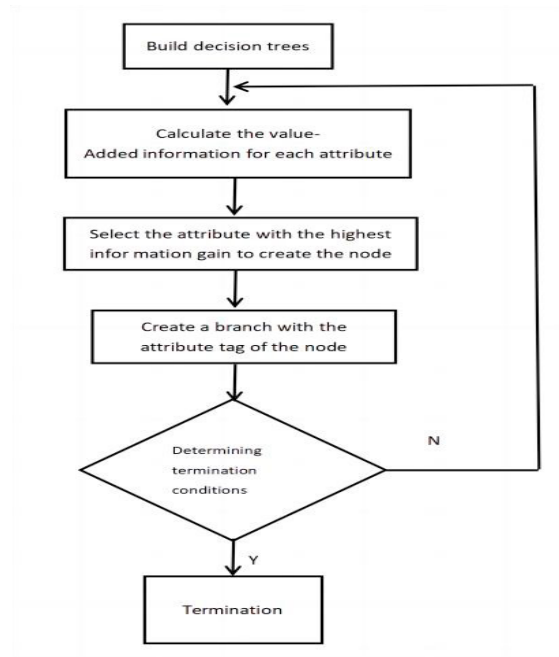
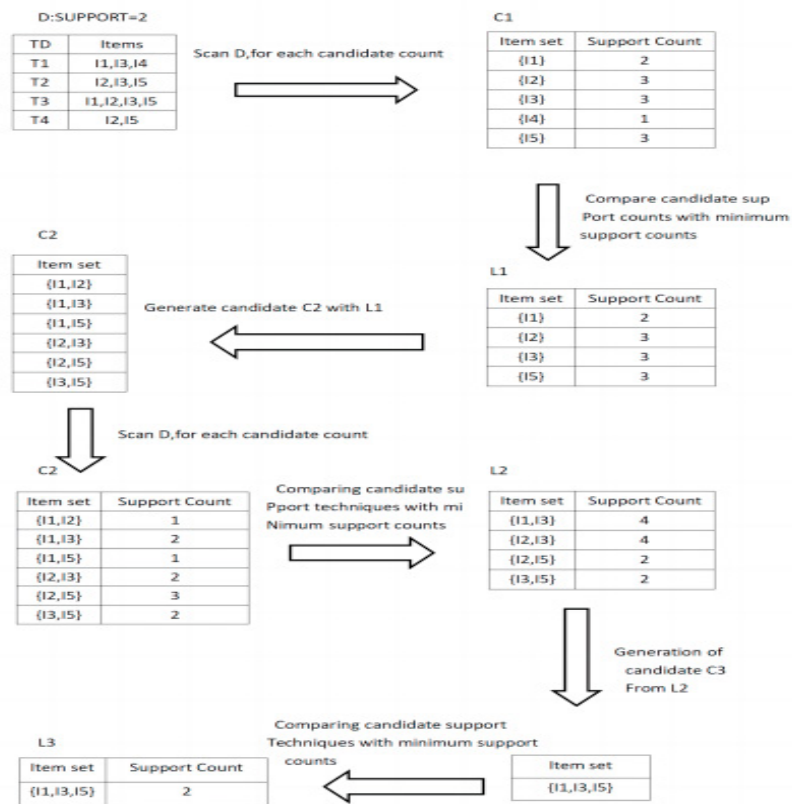


Fig. 6. Flow chart of ID3 algorithm

When users click the "Apriori Data Mining" menu item in the main interface, the association rule mining of student information will be executed, and the Apriori algorithm is shown in Figure 7. The specific code is as follows:





```

/* ---create an item set tree --- */
t = clock(); tc = 0;          /* start the timer */

apps=(char*)malloc(n *sizeof(char));

if (!apps)error(E_NOMEM);    /* get the appearance indicators */
for(apps+=i=n;--i>=0;)
    *--apps = is_getapp(itemset, i);

istree = ist_create(n, supp, conf, rsdef, apps, memopt);

if (!istree)error(E_NOMEM);  /*create an item set tree */
for(k=n;--k>=0;)            /*set single item frequencies */
    ist_setcnt(istree,k, is_getfrq(itemset,k));

ist_settac(istree, tacnt); /* set the number of transactions */

if (maxlen> maxcnt)          /* clamp the rule length */
    maxlen = maxcnt;        /* to the maximum set size */

```

**Fig. 7.** Apriori algorithm diagram

## 5. Data Mining Technology in Database Design

### 5.1 Data Integration Problems

The data integration problems of transactional databases are mainly reflected in three aspects: First, there is no uniform format standard for data extracted from various data sources, and the data are disorganized and unstable. Secondly, the data lacks uniform standards and inconsistency of the original data definition. Third, the data in the data source has changed after data integration, and the integrated data must be refreshed at a certain period.

### 5.2 Historical Data Issues

For decision support systems, comprehensive historical data is essential. Transactional databases generally store only current and short-term historical data for reasons such as the timeliness of system response. historical data, and thus it is difficult to meet these special requirements.

### 5.3 Data Processing Issues

The data needed for decision support systems is often a combination of different levels of detailed data, and transactional databases limit this combination as data redundancy.

### 5.4 Data Processing Issues

In decision analysis processing, decision support systems have changed significantly in terms of system and data requirements, which are beyond the capabilities of transactional database systems that process business in real time.

## 6. Optimization Measures Using Data Mining Techniques in Software Engineering

### 6.1. System Architecture

First, the detection of cloned code is explored, from the analysis of cloned code, the main detection method is to copy the part of the code with software engineering as the main standard. In a particular case, a part of the code is



changed according to the actual situation, and at the same time, the above code is detected and managed at the same time, and the two should be carried out at the same time, mainly to avoid the transmission of long-lived faults, leading to the failure of the whole, and the system maintenance work is realized [10]. The detection of clone codes provides security to enable the system to run smoothly [11]. Currently, there are four main methods for detecting cloned code. (1) comparison of identifiers; (2) comparison of text; (3) detection of the program structure of the system; and (4) metric circles. In the detection of cloning code, each method used has a clear diagnosis, evaluation, in the actual operation process, according to the actual situation of the cloning code detection method to make a reasonable choice. Second, the data information mining method. Data information mining method is a mining method through cross-cutting concerns, and in software engineering, the use of data information mining method can be used to transform the system. Especially in the process of information data, it can provide a variety of processing methods for data information, so that it can reach the most ideal effect. Data mining technology is widely used in various fields and has powerful functions [12].

## 6.2. *Software Development*

In the early stage, data mining technology was regarded as a database, but with the continuous development of information technology, the initial development gradually developed into real-life applications, and deeper investigation was carried out to make the system develop together with real-life applications. Software engineering, combining individual project metrics, funding, and requirements to ultimately develop a new product [13]. In traditional software engineering, it is more complicated for software development, but with the development of information technology, more and more attention is paid to software development work. The use of database mining technology in software engineering, combined with database information to make more data information with value, has promoted the development of software development [14]. In addition, software engineering reflects its own value by digging deeper into information data, making it widely available. Software engineering itself is the process of processing software development and is able to update the data in a timely manner, thus improving the quality of software development. In the current development, software development has been optimized for the previous operation process, and it is able to clearly divide the software internally, find problems and deal with them in a timely manner, and ensure that software development achieves its goals [15].

## 7. **Main Application Method**

**Association Law.** The correlation method emphasizes the internal and external connections between two things. Based on data mining techniques, it is often used for data processing and mathematical extraction. In the application of the association method, the use of relevance and interest should be reflected. In specific applications, relevant personnel need to pay attention to the connection between data mining technology and software engineering, including the support and credibility of data attributes, and use the connection of these important indicators to promote the integration of data mining technology and software engineering.

**Clustering Method.** In data mining technology, clustering is the process of dividing data objects into more detailed types and properties according to their corresponding requirements and standards. The classification principle of the clustering method is: the information data of the same type and the same attribute must maintain a high degree of similarity in the process of processing, while the information data of different attributes and different attributes must also maintain their differences in the process of processing. At present, clustering methods have been widely used in the fields of information technology and data processing. The information independent analysis feature can help software engineering to detect fraud in the commodity and its operation process, thus effectively protecting the safety and effectiveness of the operation process and improving the accuracy of information result feedback.

These methods facilitate the simplification, centralization, and efficient modeling of enterprise information.

## 8. Conclusion

The current problem is mainly the high complexity. The data generated in the current software engineering usually includes structured data types and unstructured data types, which also causes not all data mining algorithms can be used in general language software engineering, in traditional finance and other fields. The application of data mining technology in the field has formed a relatively mature result evaluation system, but the emerging software engineering data mining result evaluation system is not perfect. Due to the relatively high complexity of data information faced by software engineering developers, it is difficult to complete the evaluation of software engineering data mining results through active and effective quantitative analysis methods.

Generally speaking, the application of data mining technology in software engineering has great value. Therefore, strengthening the research of data mining technology has far-reaching significance for the overall development of software engineering. Therefore, relevant departments need to comprehensively expand the scope of research, increase the depth of research, ensure that data mining technology can be applied to software engineering more reliably and maturely, and promote its application effect to be comprehensively strengthened. With the rapid development of science and technology in our country, data mining technology has become an important way to process and effectively store information data in the information age. Data mining technology is used in software engineering to develop software and optimize the management of software engineering, thereby strengthening data mining and making data mining technology play a greater role. At the same time, while improving the quality of data mining, it is necessary to continuously shorten the time and reduce the input cost, so as to create greater benefits for my country's software engineering enterprises and provide guarantee for the sustainable development of software. engineering industry.

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